

Question 6				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
6				<u>Gene therapy techniques</u> A1 Insertion of a functional {gene/ DNA/ allele} (into cells)/ Replacement of a faulty {gene/ DNA/ allele} A2 Germ line {where (functional) gene added to embryo/ inherited} A3 Somatic gene therapy where functional gene is placed in body cells/ not inherited A4 Not permanent/needs re-treatment A5 Example of e.g.CF and CFTR gene inhaled into lungs/ Use of Restriction endonuclease/ligases/CRISPR/Cas9/stem cells/ Use of vectors/viruses/plasmids/liposomes <u>HGP link</u> B1 Aims of HGP: {sequence/ identify (position of)} {genes/ DNA} in the human genome/ improve understanding of genetic disorders/ improve the diagnosis and treatment of genetic disorders B2 {Prior to 2003/ before HGP sequenced} {very few/ fewer than 1,000} GT drug trials (per year). B3 {After 2003/ HGP sequenced} increase in GT trials B4 Reasons: (before 2003) Not all of human genome was sequenced/ (position of) {(faulty) genes/ mutations/ change in bases} was unknown/ ORA (for after 2003) B5 Up to 12,000 in 2017. <u>Use in gene therapy</u> C1 DMD is caused by a {mutation in the/ faulty} (dystrophin) gene and {failure to produce dystrophin/ protein too short/causes muscle wastage/weakness/ produces non-functional protein} C2 (Exon skipping introduces) a molecular patch over the exon with the {mutation/ fault} <i>Accept labelled diagram</i>	6	3		9		

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				C3 Complementary RNA binds to the {mutated/ fault} section of the (dystrophin) mRNA / which makes the gene readable/ transcription can take place C4 A {shorter/ different} form of {<u>dystrophin/ protein</u>} is produced C5 which is functional 7-9 marks Indicative content of this level is detail from all three sections <i>The candidate constructs an articulate, integrated account, correctly linking relevant points, such as those in the indicative content, which shows sequential reasoning. The answer fully addresses the question with no irrelevant inclusions or significant omissions. The candidate uses scientific conventions and vocabulary appropriately and accurately.</i> 4-6 marks Indicative content of this level is detail from two sections or less detail from all three sections <i>The candidate constructs an account correctly linking some relevant points, such as those in the indicative content, showing some reasoning. The answer addresses the question with some omissions. The candidate usually uses scientific conventions and vocabulary appropriately and accurately.</i> 1-3 marks Indicative content of this level is any detail from one section <i>The candidate makes some relevant points, such as those in the indicative content, showing limited reasoning. The answer addresses the question with significant omissions. The candidate has limited use of scientific conventions and vocabulary.</i>						

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				0 marks <i>The candidate does not make any attempt or give a relevant answer worthy of credit.</i>						
				Question 6 total	6	3	0	9	0	0